

Spacial Mass distributions and Galactic Stability

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Aim

- Warm Up: find a mass distribution for the trajectory:

$$z(t) = \frac{\sqrt{a^2 t^2 + 1} - 1}{a}$$

- Warm Up: Check and see if there are any other solutions for the associated lagrangian:

$$L = -\frac{m_0}{az+1} \sqrt{1 - \dot{z}^2}$$

- Main: Extend the mass distribution idea to a 2-D system and see if the mass distribution can explain Galactic Stability (the issue dark matter solves)

Dark matter has shown little experimental progress, my research aims to search for an alternative. The question I am working on is whether or not a universal non-constant mass profile could remedy the apparent motion.

To start I solved for a mass-profile (in a lagrangian) from a trajectory, then proved that the lagrangian's EOM had a single unique solution, for the time domain. Following this I will try and generalize this to two dimensions and see if this can explain galaxies low mass to angular velocity ratio.

Loschmidt's Reversibility Objection Continued

- I have just started over winter break and

End